

#29632 High CPU utilization for Angular live charts

Submitted	Received via	Requester
10 November 2023 at 16:54	Web Form	Anna Sagi <sagia@shure.com>

CCs

Alan Treadway <alan.treadway@ag-grid.com>, Alberto Gutierrez <alberto.gutierrez@ag-grid.com>

Status	Priority	Group	Assignee
Open	Normal	Support	Alan Treadway

Version	Tech Radar
v29.2.0	Tech solved

Anna Sagi 10 November 2023 at 16:54

Hello,

We are observing high CPU usage while plotting real time data, following example from AG Grid documentation.

<https://www.ag-grid.com/angular-charts/gallery/real-time-data-updates/>

I am attaching simple example (mostly copied from AG Grid documentation) where CPU reaches ~100% for 10 charts, and ~45% for 1 chart.

<https://github.com/annasagi/ag-grid-live-chart>

Please help us to resolve this issue.

Sincerely,
Anna

Alberto Gutierrez 13 November 2023 at 08:27

Hello Anna!

Thank you for your email.

An engineer is now assigned to this issue as it requires additional work. This means that we are looking to provide you with an example, or that we are looking in detail into your use case.

You will receive an update as soon as possible.

Regards,

Alberto
Head of customer service and consultancy | AG Grid

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Alberto Gutierrez 13 November 2023 at 08:28

Internal note

T2 - Do we have any opinion with regards CPU usage and charts? Does the user have a valid point?

Anna Sagi 13 November 2023 at 18:47

Hi Alberto,

Thank you for your email!

We will wait for the updates with CPU.

Meanwhile, could you please let us know if AG Grid Standalone Charts support charts synchronization ? Similar to following example (check zoom horizontally). Our goal is to build multiple charts with Time Axis X, so that we can change zoom and scrolling on all charts in time domain.

<https://www.infragistics.com/products/ignite-ui-angular/angular/components/charts/features/chart-synchronization>

Sincerely,
Anna

David Glickman 14 November 2023 at 11:08

Internal note

[Alberto Gutierrez](#)

I've linked the FR for synchronisation.

We don't really have an opinion regarding CPU usage, and I couldn't reproduce it with her example. We don't plan on looking at this in the near future.

Not sure what you want to tell her though

Alberto Gutierrez 15 November 2023 at 09:18

Hi Anna

We could not see any suspicious CPU usage on our end, so we were not able to diagnose this issue, with regards to your second point we are already tracking it in our pipeline with the following reference and description:

AG-8292

[Charts] Allow multiple chart instances to synchronise scroll, zoom, crosshairs

Note that we have no timeline for this at the moment, would you be blocked by this?

See whether this item will be in the next release by checking the **NEXT RELEASE** checkbox on the product pipeline page:

<https://www.ag-grid.com/ag-grid-pipeline/>

I would recommend signing up for ag-Grid new release notifications using [the instructions here](#). This way you'll know as soon as a new version is out and you can check whether this was implemented on [the changelog page](#).

Thanks again for bringing this up with us.

Regards,

Alberto

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Anna Sagi 15 November 2023 at 21:15

Hi Alberto,

AG-8292 i

[Charts] Allow multiple chart instances to synchronise scroll, zoom, crosshairs ----- is in backlog. We would like to try it as soon as possible. When do you think it can be prioritized ?

Regarding CPU usage, I am attaching here screenshot for 10 charts (99%) from AG Grid documentation example. For 1 chart it's about 20%. What CPU usage do you observe on your side for same example in Angular.



Best Regards,
Anna

Alberto Gutierrez 16 November 2023 at 07:20

Internal note

T2 - Do we have an ETA?

David Glickman 16 November 2023 at 10:39

Internal note

It's a big feature within Enterprise charts.
Rob says we should say that it's planned for v9.1, but no guarantees though.

Alberto Gutierrez 17 November 2023 at 13:53

Hi,

I can confirm that the dev team considers this as a high priority feature request, it is temptatively planned for

v9.1

What we would suggest is keeping an eye on the pipeline

Would this be an option?

Alberto

Head of customer service and consultancy | AG Grid

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Anna Sagi 17 November 2023 at 16:19

Hi Alberto,

I will keep an eye on the pipeline.

I have sent an attachment for high CPU usage screenshot of the example from AG-Grid official documentation (10 charts instead of one). We observe it on Windows and Mac.
Please help us to resolve the problem.

Sincerely,
Anna

Alberto Gutierrez 20 November 2023 at 08:33

Hello Anna!

We need a live code example reproducing this issue to investigate further.

There are two options to do this:

- 1- Send us a web-based reproducible scenario (plunker/codesandbox/stackblitz...)
- 2- Send us your project with instructions to run it (github / zip file...)

The most efficient way of diagnosing an issue is in order are: web-based scenario, project-based scenario and code snippets.

--- PLUNKER / CodeSandbox / WEB BASED SCENARIOS----

Please follow the steps below:

1. Locate an example in our documentation:
<https://www.ag-grid.com/vue-data-grid/integrated-charts-application-created/>
2. Select the Framework you're using from the icons on the top-right of the page
3. On the left top of the example frame, click the bottom button to open it in Plunker or CodeSandbox.
4. Update the example code so the issue is reproduced
5. Keep your code as simple as possible, only show the behavior that demonstrates the issue.
6. Save the plunker (Click the Save button in the top-left corner)
7. Send us:
 1. The link to the reproducible scenario (Copy the URL from the browser)
 2. A list of steps to follow to reproduce the issue
 3. A description of the actual(incorrect) and expected(correct) behavior

Plunker/CodeSandbox is the simplest way to provide a scenario because you can directly modify one of our documentation examples. However, if you prefer, you can use any other web-based tool such as stackblitz,

etc.

--- PROJECT-BASED SCENARIO----

If you send us your project, ensure that we can run it from the command line.

We can only investigate issues that are AG Grid-related and that have been isolated. For instance, if your issue is wrapped in your own logic, we will try to diagnose, but you might find that we come back to you asking for a simplified version of the project showing only the issue in a simpler scenario.

Looking forward to your response.

Alberto

Head of customer service and consultancy | AG Grid

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Anna Sagi 20 November 2023 at 20:02

Hi Alberto,

I have sent you a code example reproducing this issue in my original request from November 10 (you can see it in this thread).

Link to example : <https://github.com/annasagi/ag-grid-live-chart>

Instructions to run :

1. git clone <https://github.com/annasagi/ag-grid-live-chart.git>
2. open ag-grid-live-chart folder
3. npm install
4. ng serve

Chrome DevTools Performance Monitor will show ~99.5% CPU.

Regarding my second question –

In [real time plotting](#) example, only latest data is being plotted. We would like to implement similar feature allowing user to scroll or zoom to any historical data, when data is stored in time series data base.

When visible time ranges, we want to load aggregated data from server, but it seems like [AG Grid Navigator](#) doesn't allow to scroll to historical data if we change chart's data to a different range.

Visually, it should like this : [Edit fiddle - JSFiddle - Code Playground](#)

By default, we want to display last 2 minutes of data, but user should be able to zoom or scroll to any visible range.

Will it be possible to schedule a meeting for more details, if it's still unclear ?

Sincerely,
Anna

Alberto Gutierrez 21 November 2023 at 08:49

Hello Anna!

Thank you for your email.

An engineer is now assigned to this issue as it requires additional work. This means that we are looking to provide you with an example, or that we are looking in detail into your use case.

You will receive an update as soon as possible.

Regards,

Alberto
Head of customer service and consultancy | AG Grid

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Alberto Gutierrez 21 November 2023 at 08:50

Internal note

T2 - User insists on us looking at the CPU usage from their repro... Would it be possible to get back to them with some more information?

David Glickman 21 November 2023 at 09:07

Internal note

[Alberto Gutierrez](#) - can you reproduce?
I took a look last time and didn't see any dramatic cpu usage.

Alberto Gutierrez 22 November 2023 at 09:19

Internal note

I have attached my results, I think a dev might want to let us know if this is to be expected

David Glickman 23 November 2023 at 15:00

Internal note

[Alberto Gutierrez](#) we won't be able to look at this until the end of next week most probably

Anna Sagi 29 November 2023 at 00:18

Hello Alberto,

Could you please give us update regarding this case ?

Sincerely,
Anna

Alberto Gutierrez 29 November 2023 at 08:43

Hi Anna,

This ticket has been raised to the dev team, but at the moment we are in the middle of releasing v31, which is taking up most of our resources, as soon as this release is out, which should be in the next few days, they will be able to look into this.

Does this make sense?

Alberto
Head of customer service and consultancy | AG Grid

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Alberto Gutierrez 30 November 2023 at 12:49

Internal note

T2 - Would there be capacity now to look at this?

David Glickman 30 November 2023 at 13:53

Internal note

[Alan Treadway](#) - can you take a look at this?

Link to example : <https://github.com/annasagi/ag-grid-live-chart>

Instructions to run :

1. git clone <https://github.com/annasagi/ag-grid-live-chart.git>
2. open ag-grid-live-chart folder
3. npm install
4. ng serve

Customer claims that Chrome DevTools Performance Monitor will show ~99.5% CPU.
I don't see that on my machine.

Question is if the CPU usage is 'high' or 'normal' and if we can/will do anything about it.

David Glickman 4 December 2023 at 16:11

Hello Anna,

Thank you for the context.

We can see there is high CPU usage in your example. This is likely caused by the frequency of your data updates - at 50ms intervals. Our charts are not optimised for this high frequency of updates.

However, we agree this would be a valuable addition to the product and we have added feature requests to improve our performance in this scenario.

We have added this requirement to our backlog and we are tracking it with the following reference and description:

AG-10034 [Charts] Performance improvements for updating Axis scales

AG-10035 [Charts] Improve performance with high frequency updates

We have no timeline for this at the moment, but this can still be picked up by our team when they're making modifications to this functional area of AG Grid. Please use the functionality currently provided by AG Grid until this is delivered at a later point.

See whether this item will be in the next release by checking the **NEXT RELEASE** checkbox on the product pipeline page:

<https://www.ag-grid.com/ag-grid-pipeline/>

The best way to track this is to sign up for AG Grid new release notifications using [the instructions here](#). This way you'll know as soon as a new version is out and you can check whether this specific item was implemented on [the changelog page](#).

Thanks again for bringing this up with us.

Regards,

David
Technical Product Analyst | AG Grid

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